Keaton Van Patten

AI Systems Engineer

Rancho Cordova, CA · remote or Sacramento area · keatonvp123@gmail.com · github.com/Peacho97 · peacho97.github.io · linkedin.com/in/keaton-van-patten-227640272

SUMMARY

Ships production AI systems end to end, then measures what they actually do. Sole developer of a 200K-line agentic platform (678 API endpoints, 122-tool agent, ~3,000 tests) and of a live scheduling system a healthcare clinic uses every day. Architecture decisions here trace back to numbers measured on real hardware, including the numbers that proved earlier claims wrong.

SELECTED PRODUCTION SYSTEMS

Clinic Scheduling System + Outage Failover *- pediatric therapy clinic (live since Aug 2026)*

- Sole developer of a workload-based scheduling system that replaced a fully paper process: ~35K lines, 30 REST endpoints, 15 role-gated pages, written in PHP 8.3 with SQLite and a zero-dependency JS PWA. Clinic staff use it daily, 573 audited actions in the first 18 days.
- Designed an 8-component weekly workload model that computes therapist utilization to the quarter-hour and hard-blocks anything over capacity. A filter-and-rank substitute selector covers absences.
- Found a real production outage, then built and deployed a zero-dependency Node failover mailer on Fly.io in under 48 hours. It is outbound-only with no public IP, and it minimizes PHI server-side so only de-identified case codes ever leave the host. 157 tests cover it, including both DST transitions.
- Put the application through an in-house security-review desk before launch and fixed the two critical findings it returned, an install-takeover path and cross-role data exposure. Shipped Argon2id auth, email OTP 2FA, CSRF hardening, and a 14-action audit log. Wrote 800+ integration test assertions with no third-party test framework.

RAG Reference Service ("Librarian") *- production retrieval over a 19K-document corpus*

- Built and operate a retrieval-augmented Q&A service over 3,190 files, 14,534 code symbols and 18,094 embedding vectors. Okapi BM25 and dense-vector retrieval run side by side, get fused with Reciprocal Rank Fusion, then reranked. Incremental content-hash embedding made batch runs 21.8× faster, measured.
- Serves AI agents as an 8-tool MCP server, 613 live queries in 30 days with 67% of them machine-to-machine, behind a 5-provider LLM fallback chain that answers ~92% of calls on the low-cost primary.
- Built a CI-gated evaluation harness of 22 graded questions. A release is blocked when graded accuracy drops under 90%, and the deterministic questions have to pass with every model provider offline. Measured accuracy went from 86.4% to 100%.

Autonomous Incident Desk *- three-tier deflection ladder*

- Built a self-healing service desk that watches 15 production backends on a 15-second probe loop. Over 53 days it took 612 incidents, auto-resolved 72.9% of them at a 55-second mean time-to-resolution, and escalated 0.33% to a human.
- Deflection is cost-tiered. Deterministic playbooks run first and settle 75.6% on the first line, LLM triage only runs on the 3.3% of tickets they miss, and the model contract is diagnosis-only. Circuit breakers and a kill-pattern validator bound the blast radius.
- These are infrastructure incidents rather than customer-support tickets, so the deflection architecture is the part that transfers, not the domain.

Multi-Provider LLM Platform *- cost governance and orchestration*

- Operate a metered fleet of 21 models across 8 providers including Anthropic, DeepSeek and local Ollama, with a USD-exact ledger entry for every call. So far that covers 2,129 calls, 41M tokens, 38 attributed use-case surfaces, and a 64% prompt-cache hit rate.
- Cut cost per model call 71% while volume grew 51%. Profiled provider pricing, latency and data sensitivity, moved background workloads onto cheaper tiers, and pinned latency-critical and private-data paths to local GPUs.
- Stood up a self-hosted LiteLLM control plane on PostgreSQL that issues per-use-case virtual API keys with hard monthly budgets. A raw-socket reachability probe switches traffic from the gateway to direct provider calls automatically, and 852 calls have been served uninterrupted through that fallback path.

Personal AI Platform ("Companion Hub") *- 122-tool production agent*

- Built a local-first agentic platform in Python/FastAPI with a Svelte/Electron front end, 678 HTTP endpoints and ~3,000 tests. Its agent dispatches 122 real tools across filesystem, shell, calendar, health and messaging, all behind a three-layer authorization model of fail-closed per-surface allowlists, chokepoint denylists and a credential-aware filesystem perimeter, with 32 dedicated gate tests.
- Designed a token-budgeted 5-zone prompt architecture that keeps the static prefix byte-identical across turns so the inference cache hits. A curated tool subset plus scored tool search cut small-model tool-catalog overhead 5.3×, from 7.4K tokens to 1.4K.
- Built an AI security-review desk with 110 consultations, 165 specialist reviews and 548 tracked checklist items. Local triage models were benchmarked adversarially before selection; the one chosen resisted 6/6 prompt injections, and a candidate was rejected for obeying injected verdicts.
- Run dual-GPU local inference infrastructure. A cross-process GPU lease broker with priority aging and fencing tokens fails over in a measured 73 seconds, video generation runs on AMD ROCm at 87% VRAM utilization, and a 744B-parameter MoE model was tuned +86% on consumer hardware.

EXPERIENCE

HRBC Inc. - Systems Manager & Software Developer · Sacramento, CA Sept 2020 – Present

- Sole technical owner for a mortgage brokerage, covering systems administration, Microsoft 365 and Entra ID, user access management, vendor systems such as LOS, CRM and pricing engines, incident response, and training for loan officers. Five years of requirements discovery done directly with non-technical stakeholders.
- Built an internal operations hub with an embedded AI assistant, running a hand-rolled agentic tool-use loop on the Anthropic API with 8 typed JSON-schema tools, bounded round-trips and mutation tracking over a 15-system vendor registry. Security policy lives in the prompt and the tool schemas, and the registry stores where credentials are kept, never the secrets themselves. The agentic backend is complete and demonstrated via API; the UI was never wired up.
- Originated loans and advised clients directly; trained and supervised loan officers and assistants.

MP NexLevel - Utility Technician 2018 – 2020

Crew lead on utility infrastructure projects; equipment operation and maintenance.

SKILLS

AI/LLM: agent orchestration · multi-agent systems · RAG (BM25, vector embeddings, RRF hybrid search) · tool calling / function calling · MCP (Model Context Protocol) · prompt engineering · LLM evaluation harnesses · LLM-as-judge · guardrails & prompt-injection defense · human-in-the-loop gating · cost attribution & model routing · local inference (Ollama, llama.cpp, CUDA, ROCm) · Claude API · OpenAI-compatible APIs · DeepSeek · Whisper STT · TTS

Engineering: Python · FastAPI · TypeScript/JavaScript · SQL/SQLite/PostgreSQL · Svelte · Electron · REST API design · WebSockets/SSE · PHP · Docker · CI/CD (GitHub Actions) · pytest · Fly.io · Tailscale · Windows/Linux administration

Customer-facing: requirements discovery · stakeholder management · technical training · incident communication · security-review facilitation

EDUCATION

Google IT Support Certification · Google Computer Networking Certification